(i) $\int_a^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_a^{t} f(x)\,dx$

(ii) $\int_{-\infty}^{a} f(x)\,dx = \lim_{t\to-\infty} \int_t^{a} f(x)\,dx$

(iii) $\int_{-\infty}^{\infty} f(x)\,dx = \lim_{t\to\infty} \int_{-t}^{t} f(x)\,dx$

They are called **improper integrals of first kind. If the limits exist, then the improper integrals are said to be convergent.**

Note

By the Fundamental theorem of integral calculus, there exists a function $F(t)$ such that

$$\int_a^t f(x)\,dx = F(t)-F(a)$$

$$\therefore \int_a^{\infty} f(x)\,dx = \lim_{t\to\infty}\int_a^t f(x)\,dx = \lim_{t\to\infty}[F(t)-F(a)] = \left[\int f(x)\,dx\right]_a^{\infty}.$$

Example 9.35

Evaluate $\int_b^{\infty} \frac{1}{a^2+x^2}\,dx$, $a>0, b\in\mathbb{R}$.

Solution

We have $\int_b^{\infty} \frac{1}{a^2+x^2}\,dx = \left[\frac{1}{a}\tan^{-1}\frac{x}{a}\right]_b^{\infty} = \frac{1}{a}\tan^{-1}\infty - \frac{1}{a}\tan^{-1}\frac{b}{a} = \frac{1}{a}\left[\frac{\pi}{2}-\tan^{-1}\frac{b}{a}\right]$.

Note

From the above example, we get

(i) $\int_0^{\infty} \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\left[\frac{\pi}{2}-\tan^{-1}0\right] = \frac{\pi}{2a}$.

(ii) $\int_a^{\infty} \frac{1}{a^2+x^2}\,dx = \frac{1}{a}\left[\frac{\pi}{2}-\tan^{-1}1\right] = \frac{1}{a}\left[\frac{\pi}{2}-\frac{\pi}{4}\right] = \frac{\pi}{4a}$.

(iii) $\int_{-\infty}^{\infty} \frac{1}{a^2+x^2}\,dx = \lim_{t\to\infty}\int_{-t}^{t} \frac{1}{a^2+x^2}\,dx = \lim_{t\to\infty} 2\int_0^t \frac{1}{a^2+x^2}\,dx$, since $\frac{1}{a^2+x^2}$ is even function

$$= 2\int_0^{\infty} \frac{1}{a^2+x^2}\,dx = 2\left(\frac{\pi}{2a}\right) = \frac{\pi}{a}.$$

Example 9.36

Evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{4\sin^2 x+5\cos^2 x}$.

Solution

Let $I = \int_0^{\frac{\pi}{2}} \frac{dx}{4x\sin^2 x+5\cos^2 x}$

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